Research on Chatbots in Various Domains

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1. Education

Pounce (Georgia State University)

- **Scenario:**Admission support,onboarding and administrative Q&A via SMS.
- **Implementation:** Custom AI chatbot platform,integrated with Georgia State systems, sending personalized reminders and automatically answering support queries.
- **Impact:** Tens of thousands of messages sent to students; increased retention and graduation rates, efficient case deflection.
- **Sources:**
  - `https://news.gsu.edu/2022/03/21/classroom-chatbot-improves-student-performance-study-says/`
  - `https://mainstay.com/case-study/how-georgia-state-university-supports-every-student-with-perso`
  - `https://completega.org/sites/default/files/meetings/MomentumIV/GSU_MSIV_Chatbots.pdf`
- **What we learn for university chatbots:**
  - Automated reminders and tailored onboarding support greatly improve student engagement and success. Integration with university systems helps provide personalized, relevant information at scale.

2. Healthcare

Babylon Health

- **Scenario:** Symptom checker chatbot for preliminary medical advice and triage.
- **Implementation:** AI + NLP, mobile and web app, combining rule-based and machine learning approaches.
- **Critical Notes:** Used widely in the UK and globallypeer-reviewed studies raised some questions about accuracy compared to human doctors. Ceased US operations in 2023.
- **Sources:**
  - `https://en.wikipedia.org/wiki/Babylon_Health`
  - `https://static.ie.edu/CGC/Case-Study-1-Babylon-Health.-IE-CGC.pdf`
- **What we learn for university chatbots:**
  - Chatbots can effectively handle repetitive or urgent questions and triage requests, reducing the load on support staff. A university chatbot can act as a first responder for common issues or direct users to the right office.

3. Finance / Banking

Erica (Bank of America)

- **Scenario:** Virtual financial assistant for account management, spending tracking, payments, and financial advice.
- **Implementation:** Internally developed using NLP and ML, fully integrated in Bank of America's mobile app.
- **Impact:** Used by more than 42 million customers, with over 2 billion interactions as of 2024.
- **Sources:**
  - `https://info.bankofamerica.com/en/digital-banking/erica`
  - `https://www.reuters.com/business/finance/bofa-says-its-virtual-financial-assistant-used-more-t`
  - `https://tearsheet.co/podcasts/the-story-of-erica-bank-of-americas-homegrown-digital-assistant/`
- **What we learn for university chatbots:**
  - Deep integration with existing digital services is key. A university chatbot should access real student data, timetables, and platforms to offer relevant, personalized help while also learning and improving over time.

4. Administrative / Customer Support

Intercom

- **Scenario:** Customer service automation via chatbot, including ticket deflection, live chat, and help center integration.
- **Implementation:** Intercom's Fin AI Agent resolves up to 50–60% of support queries autonomously and can escalate to human agents when needed
- **Features:** Live chat, automated response flows, FAQ handling, ticket classification and routing, knowledge-base integration, self-service article suggestions
- **Sources:**
  - `https://www.intercom.com/learning-center/conversational-ai-for-customer-service`
  - `https://www.intercom.com/learning-center/automated-ticketing-system`
  - `https://www.intercom.com/suite`
- **What we learn for university chatbots:**
  - Availability 24/7 and the ability to escalate complex issues to human staff are essential. For a university, a chatbot should handle repetitive inquiries and provide instant help, but also offer an easy way for students to connect to real people when needed.

5. Other Notable Examples

Woebot (Mental Health)

- **Scenario:** Emotional support and conversation-based therapy for mental health.
- **Implementation:** Empathetic, conversational chatbot modeled as a digital therapist.
- **Sources:**
  - `https://www.woebothealth.com/`
  - `https://www.gadgethacks.com/news/your-new-shrink-is-only-facebook-message-away-with-stanfords-`
- **What we learn for university chatbots:**
  - Communication style and empathy are important, even for automated systems. A university chatbot that uses a friendly, supportive tone can help students feel more comfortable, especially when discussing sensitive topics.

References

- Georgia State University: Use of the Pounce chatbot for student communication, onboarding, and retention.
- Babylon Health: Symptom checker chatbot, global deployment and critical reviews.
- Bank of America: The Erica virtual assistant, mobile integration.
- Woebot: Mental health conversational chatbot, digital therapist model.
- Intercom: Industry-wide adoption of AI chatbots for customer support.

6. Use Cases and Relationships

6.1 Guest Use Case

Guest Use Case

- **Guides**
  - **Website Navigation** – Get assistance with navigating the university's website.
  - **Bachelor Application** – Access guidance and procedures for applying to bachelor's degree programs.
  - **Master Application** – Find information and steps for applying to master's degree programs.
  - **Administrative Procedures** – Understand various administrative processes and requirements.
  - **Research** – Explore research activities, projects, and collaborations.
  - **Employment** – Discover job opportunities or career support services offered through the university.
- **Actions**
  - **Login / Create Account** – Option for guests to register or sign into a personal account.
  - **View Group Student List** – Faculty office staff can access lists of students within specific academic groups or cohorts.
- **Other**
  - **Study Program Recommendations** – The system suggests programs that align with the user's interests.
  - **Specialization Choice Simulation** – Lets users experiment with different specialization paths.
  - **Q&A** – Basic question-answer interface for general queries.
  - **Chit chat** – Casual interaction with the system to enhance user engagement.
- **Information**
  - **Leadership** – Learn about the university's top management and their roles.
  - **University** – General information about the university's history, mission, and structure.
  - **Contact** – How to get in touch with different university departments or support services.
  - **International Relations** – Overview of international programs, student exchanges, and global partnerships.
  - **Departments** – Basic info about each department and their focus areas.
  - **Staff** – View key academic or administrative staff involved in university operations.
  - **Campus Location** – Maps and directions to university campuses and buildings.
  - **Course Schedule** – View the timetable and details for enrolled courses.
  - **Exam Schedule** – Access information regarding examination dates, times, and locations.
  - **Make-up Exam Schedule** – Check schedules for make-up examinations.
  - **Mobilities** – Find details about international exchange programs or student mobility opportunities.
  - **Groups & Students List** – View lists of students within specific academic groups.
  - **Student Organizations** – Explore information about various student organizations and clubs.
  - **Contact** – Find contact details for university departments or support services.
  - **Required Documents Details** – Access information about necessary documents for various university processes.
  - **Tuition Payments** – View details related to tuition fees and payment status.

1 LLMs

In this section, we analyze some of the leading open-source Large Language Models (LLMs) suitable for building an AI chatbot for university use: **Meta Llama-3-Chat**, **Mistral (Mixtral-8x7B Instruct)**, and **Qwen1.5-Chat**.

1.1 Meta Llama-3-Chat

Meta Llama-3-Chat is Meta's flagship open-source LLM, optimized for conversational AI and instruction-following tasks. It is available in multiple sizes (8B and 70B parameters) and supports multilingual input, making it highly relevant for diverse university environments.

Advantages:

- State-of-the-art performance for open LLMs, rivals GPT-4 on many benchmarks.
- Excellent conversation flow, context retention, and follow-up handling.
- Multilingual, supporting complex international campus needs.
- Long context window (up to 128k tokens), suitable for document-based Q&A.
- Strong community and frequent updates.
- Runs fully on-premises for maximum privacy.

Disadvantages:

- Requires modern GPU hardware (8B: ∼12GB VRAM; 70B: server-class GPUs).
- Large disk and memory footprint.
- License is mostly permissive but should be reviewed for commercial projects.

Usefulness in our context: Llama-3-Chat is a top choice for robust, multilingual university chatbots that need both privacy and advanced conversation quality.

1.2 Mistral (Mixtral-8x7B Instruct)

Mixtral-8x7B Instruct is a Mixture-of-Experts (MoE) model developed by Mistral AI. It combines high efficiency with strong performance, using a unique sparse architecture to activate only parts of the model per query.

Advantages:

- Apache 2.0 license: fully open for all uses.
- Outperforms most 13–30B models with only ∼13B active parameters.
- Efficient and fast ideal for real-time chatbot deployments.
- Handles long conversations and large context windows.
- Lower hardware requirements than Llama-3-70B.

Disadvantages:

- Slightly less nuanced than Llama-3-70B on some complex or creative tasks.
- Multilingual support is good, but not as advanced as Qwen1.5.
- Still needs a modern GPU (about 13GB VRAM for quantized models).

Usefulness in our context: Mistral is excellent for efficient, real-time campus chatbots where hardware resources are moderate and open licensing is important.

1.3 Qwen1.5-Chat

Qwen1.5-Chat is an open-source chat-optimized LLM developed by Alibaba. It is known for exceptional multilingual support (35+ languages), large context windows, and high leaderboard rankings.

Advantages:

- Best-in-class multilingual support among open LLMs.
- Very long context windows (up to 32k tokens), suitable for complex Q&A.
- High performance in conversational tasks; competitive with larger models.
- Flexible open-source licensing.

Disadvantages:

- Community and documentation are growing but less established than Llama/Mistral.
- Larger models require substantial hardware (7B manageable, 72B server-class).
- Primary support and updates are from Alibaba.

Usefulness in our context: Qwen1.5-Chat is ideal for universities with diverse, international populations, or projects needing advanced language coverage.

1.4 Command R+ (Cohere)

Command R+ is a specialized retrieval-augmented generation (RAG)-optimized model developed by Cohere. It excels at fast, accurate information retrieval, summarization, and structured query handling, especially in enterprise and academic knowledge management systems.

Advantages:

- Best-in-class RAG performance among open-access or commercial LLMs.
- Optimized for document summarization, and indexing large content.
- Fast and lightweight inference—efficient in both cloud and local deployments.
- Strong candidate for internal university knowledge assistants or student portals.

Disadvantages:

- Less fluent in creative writing or general conversation compared to Claude or Gemini.
- Coding and logic skills are average—not ideal for technical tutoring.
- Supports major languages but not as extensive as Qwen1.5 or LLaMA-3
- No multimodal support.
- Smaller ecosystem and community than Meta or Mistral.

Usefulness in our context: Command R+ is ideal for retrieval-heavy university tools, such as library search assistants, policy explainers, or admin-facing chat tools.

1.5 Comparison Table

Model	**Strengths**	**Limitations**	**Best Use**
Llama-3-Chat	Top conversation, multilingual, privacy	Needs strong GPU, large size	Advanced, private, robust bots
Mixtral-8x7B	Efficient, fast, open license	Less nuance, moderate GPU	Real-time, scalable chatbots
Qwen1.5-Chat	Multilingual, long context, flexible	Growing docs, hardware for large models	International/multilingual bots
Command R+	Top retrieval-augmented tasks, efficient, open base model	Limited creativity, no multimodal, smaller ecosystem	Document Q&A, knowledge assistants

Table 1: Comparison of LLMs for university chatbot applications

2 Integrating LLMs into a Website

When deploying a university chatbot powered by a Large Language Model, several architectural methods are available for integrating the model with the website frontend.

2.1 Direct API Integration (Cloud or Hosted LLM)

In this approach, the website (frontend or backend) communicates with a remote LLM service through an API (such as OpenAI, Hugging Face Inference Endpoints, or a third-party SaaS). User questions are sent to the LLM as API requests, and responses are displayed on the website.

Advantages:

- Quick to deploy, with minimal setup and no need for local infrastructure.
- Access to the latest, most powerful models.
- Scalable to large numbers of users.

Disadvantages:

- Ongoing costs, typically priced per token or request.
- User data is processed by external servers, raising privacy concerns.
- Limited control over customization or fine-tuning the LLM.

Usefulness in our context: Suitable for quick prototyping or where the university does not have access to powerful hardware. (not ideal in our situation and probably won't be used)

2.2 Self-hosted or Local LLM Backend

Here, the university runs the LLM on its own infrastructure (using tools such as Ollama, Hugging Face Transformers). The website communicates with the LLM via a REST API on the university's server.

Advantages:

- Full data privacy and security, as all information remains on university servers.
- No usage-based costs, unlimited queries after deployment.
- Full control over model updates, customization, and integrations.

Disadvantages:

- Requires access to GPU-equipped servers and technical expertise.
- More complex setup, deployment, and maintenance.
- Initial load and inference may be slower on limited hardware.

Usefulness in our context: Recommended for production deployments where control is essential, and where university IT resources are available.

2.3 In-browser LLM Inference

This method runs the LLM directly in the user's browser using JavaScript (e.g., transformers.js or WebLLM). The model is downloaded to the client, and all inference is performed locally.

Advantages:

- Ultimate privacy: user data never leaves their device.
- No server infrastructure or cloud costs.
- Works offline after the model loads.

Disadvantages:

- Only feasible for very small models due to browser and memory constraints.
- Large initial download time for model weights.
- Lower response quality and speed compared to cloud or server-hosted LLMs.

Usefulness in our context: Best for demos, educational tools, or privacy-critical scenarios where only lightweight responses are needed.

2.4 Hybrid Integration

A hybrid approach may combine the above methods: for example, using a cloud API as the main backend, with a local or in-browser model as a fallback for privacy or offline use.

Advantages:

- Flexibility to balance privacy, cost, and performance.
- Robustness: continued functionality if one backend is unavailable.

Disadvantages:

- More complex implementation and deployment logic.
- Potential for inconsistent user experience.

Usefulness in our context: Can provide an optimal user experience for a university chatbot serving diverse needs and device capabilities.

2.5 Comparison Table

Method	**Pros**	**Cons**	**Best For**
Cloud API	Fast, easy, scalable	Cost, privacy, less control	Rapid prototyping, public info bots
Self-hosted	Privacy, no cost, full control	Needs server/IT, complex	Production, sensitive data
In-browser	Max privacy, offline, no infra	Only tiny models, slow, large download	Demos, high privacy, small tasks
Hybrid	Flexible, robust	Complex logic	Diverse/large-scale deployments

Table 2: Comparison of LLM integration methods for university chatbots

2.6 Conclusion

Given the university's ability to manage its own infrastructure, the **self-hosted (local) LLM integration** method is the optimal choice for our project. This approach ensures that all user interactions and information remain securely within the university's servers, eliminates usage-based costs, and allows for maximum customization and control over the chatbot's capabilities.